MAT 344 HW08

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Problem 1 (p.178#2)**.** Show that the series

$$\zeta(z) = \sum_{n=1}^{\infty} n^{-z}$$

converges for $\operatorname{Re} z > 1$, and represent its derivative in series form.

Proof. For any $c > 1$, if $\operatorname{Re} z \geq c$, then $|n^{-z}| \leq n^{-c}$. Since $\sum_{n=1}^{\infty} n^{-c} < \infty$, the Weierstrass M-test implies that $\zeta(z)$ converges uniformly on $\mathbb{H}_c = \{z \in \mathbb{C} : \operatorname{Re} z \geq c\}$. Since any compact subset of this domain of convergence is contained in $\mathbb{H}_c$ for some c, Theorem 1 on p.176 implies that the derivative of ζ can be taken termwise, so

$$\boxed{\zeta'(z) = \sum_{n=1}^{\infty} -n^{-z} \log n.}$$

□

Problem 2 (p.190#1)**.** Comparing coefficients in the Laurent developments of $\cot(\pi z)$ and of its expression as a sum of partial fractions, find the values of

$$\sum_{n=1}^{\infty} \frac{1}{n^2}, \qquad \sum_{n=1}^{\infty} \frac{1}{n^4}, \qquad \sum_{n=1}^{\infty} \frac{1}{n^6}.$$

Solution. Equation (10) on p.189 says that

$$\pi \cot(\pi z) = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2} = \frac{1}{z} - \sum_{n=1}^{\infty} \frac{1}{n^2} \cdot \frac{2z}{1 - (\frac{z}{n})^2} = \frac{1}{z} - \sum_{n=1}^{\infty} \sum_{k=0}^{\infty} \frac{2z}{n^2} \cdot \frac{z^{2k}}{n^{2k}} = \frac{1}{z} - \sum_{k=1}^{\infty} 2\zeta(2k) z^{2k-1}.$$

On the other hand, $\pi \cot(\pi z) = \frac{\pi \cos(\pi z)}{\sin(\pi z)}$, so the Laurent series for $\pi \cot(\pi z)$ around zero is given by

$$\begin{aligned}
\pi \cdot \frac{1 - \frac{(\pi z)^2}{2!} + \frac{(\pi z)^4}{4!} - \cdots}{\pi z - \frac{(\pi z)^3}{3!} + \frac{(\pi z)^5}{5!} - \cdots} &= \frac{1}{z} \cdot \frac{1 - \frac{(\pi z)^2}{2!} + \frac{(\pi z)^4}{4!} - \cdots}{1 - \frac{(\pi z)^2}{3!} + \frac{(\pi z)^4}{5!} - \cdots} \\
&= \frac{1}{z} \left(1 - \frac{(\pi z)^2}{2!} + \frac{(\pi z)^4}{4!} - \frac{(\pi z)^6}{6!} \cdots \right) \left(1 + \frac{\pi^2 z^2}{6} + \frac{7\pi^4 z^4}{360} + \frac{31\pi^6 z^6}{15120} + \cdots \right).
\end{aligned}$$

Comparing coefficients between the the two equations tells us that

$$\begin{aligned}
-2\zeta(2) &= \frac{-\pi^2}{2} + \frac{\pi^2}{6} = -\frac{\pi^2}{3} \implies = \boxed{\zeta(2) = \frac{\pi^2}{6}} \\
-2\zeta(4) &= -\frac{\pi^4}{12} + \frac{7\pi^4}{360} + \frac{\pi^4}{24} = -\frac{\pi^4}{45} \implies \boxed{\zeta(4) = \frac{\pi^4}{90}} \\
-2\zeta(6) &= \frac{31\pi^6}{15120} - \frac{-7}{720} + \frac{1}{144} - \frac{1}{720} = -\frac{2\pi^6}{945} \implies \boxed{\zeta(6) = \frac{\pi^6}{945}}
\end{aligned}$$

□

Problem 3 (p.193#1)**.** Show that

$$\prod_{n=2}^{\infty}\left(1-\frac{1}{n^2}\right)=\frac{1}{2}.$$

Solution. This follows from

$$\lim_{N\to\infty}\prod_{n=2}^{N}\left(1-\frac{1}{n^2}\right)=\lim_{N\to\infty}\prod_{n=2}^{N}\frac{(n+1)(n-1)}{n^2}=\lim_{N\to\infty}\frac{1}{2}\cdot\frac{N+1}{N}=\boxed{\frac{1}{2}}$$

where the second-to-last equality is true because for $2<n<N$, the n^2 term in the denominator is canceled by the two ns in the numerators of the previous and next terms of the product. $\square$

Problem 4 (p.193#2)**.** Prove that for $|z|<1$,

$$(1+z)(1+z^2)(1+z^4)(1+z^8)\cdots=\frac{1}{1-z}.$$

Solution. Observe that the product of the first N terms on the left-hand side are equal to $\frac{1-z^{2^N}}{1-z}$ by difference of squares; clearly this is true for $N=1$, and

$$\frac{1-z^{2^N}}{1-z}\cdot(1+z^{2^N})=\frac{1-z^{2^{N+1}}}{1-z},$$

so the claim is true for all N. Then $z^{2^{N+1}}$ goes to zero when N grows arbitrarily large because $|z|<1$, so

$$\prod_{n=0}^{\infty}(1+z^{2^n})=\lim_{N\to\infty}\prod_{n=0}^{N}(1+z^{2^n})=\lim_{N\to\infty}\frac{1-z^{2^{N+1}}}{1-z}=\boxed{\frac{1}{1-z}}. \qquad \square$$

Problem 5 (p.193#3)**.** Prove that

$$\prod_{n=1}^{\infty}\left(1+\frac{z}{n}\right)e^{-z/n}$$

converges absolutely and uniformly on every compact set.

Solution. Expanding $e^{-z/n}$ as a power series and multiplying, we get

$$\prod_{n=1}^{\infty}\left(1+\sum_{k=2}^{\infty}(-1)^k\left(\frac{1}{(k-1)!}-\frac{1}{k!}\right)\left(\frac{z}{n}\right)^k\right),$$

which converges absolutely and uniformly on a given compact set K if and only if the sum

$$\sum_{n=1}^{\infty}\left(\sum_{k=2}^{\infty}(-1)^k\left(\frac{1}{(k-1)!}-\frac{1}{k!}\right)\left(\frac{z}{n}\right)^k\right)$$

also converges absolutely and uniformly on K.

Since K is compact, it is bounded, say $|z|<N$ for some integer N whenever $z\in K$. Discarding finitely many terms in the series under consideration does not affect the nature of its convergence, so we consider the sum starting at $n=N$ instead. Then, we get both absolute and uniform convergence by the following chain of inequalities:

$$\begin{aligned}\sum_{n=1}^{\infty}\left|\sum_{k=2}^{\infty}(-1)^k\left(\frac{1}{(k-1)!}-\frac{1}{k!}\right)\left(\frac{z}{n}\right)^k\right| &\le \sum_{n=N}^{\infty}\sum_{k=2}^{\infty}\left(\frac{1}{(k-1)!}-\frac{1}{k!}\right)\frac{|z|^k}{n^k}\\ &\le \sum_{n=N}^{\infty}\sum_{k=2}^{\infty}\left(\frac{1}{(k-1)!}-\frac{1}{k!}\right)\frac{N^2}{n^2}=\sum_{n=N}^{\infty}\frac{N^2}{n^2}<\infty.\end{aligned} \qquad \square$$

Problem 6 (p.197#1)**.** Suppose that $a_n \to \infty$ and that the A_n are arbitrary complex numbers. Show that there exists an entire function $f(z)$ which satisfies $f(a_n) = A_n$.

Solution. Let $g(z)$ have simple zeroes at each a_n and consider

$$f(z) = \sum_{n=1}^{\infty} g(z) \frac{e^{\gamma_n(z-a_n)}}{z-a_n} \cdot \frac{A_n}{g'(a_n)}.$$

for some constants γ_n yet to be determined. Certainly if the sum converges locally uniformly then $\lim_{z\to a_n} f(z)$ is A_n, since

$$\lim_{z\to a_n} \frac{g(z)}{(z-a_n)g'(a_n)} \cdot e^{\gamma_n(z-a_n)} = \lim_{z\to a_n} \frac{g(z)-g(a_n)}{(z-a_n)g'(a_n)} \cdot e^{\gamma_n(z-a_n)} = 1$$

while all the other addends go to zero because the factors excluding $g(z)$ would be bounded (using uniform convergence to swap sum and limit).

not sure how to pick γ_n :(□

Problem 7 (p.198#3)**.** What is the genus of $\cos(\sqrt{z})$?

Solution. Since $\cos(z)$ is a translate of $\sin(z)$, it has the same genus as $\sin(z)$, so

$$\cos(z) = e^{g(z)} \prod_{n\in\mathbb{Z}} \left(1 - \frac{2z}{\pi(2n-1)}\right) e^{2z/(\pi(2n-1))}.$$

The same strategy as the one employed by Ahlfors to conclude that the product converges uniformly and that $g(z)$ is a constant in the product expansion for $\sin(z)$ also can be used here to show that $g(z)$ is constant here. Pairing factors of the product with positive and negative coefficients shows that

$$\cos(z) = C \cdot \prod_{n=1}^{\infty} \left(1 - \frac{4z^2}{\pi^2(2n-1)^2}\right)$$

where the $e^{\bullet}$ terms cancel each other in each pairing. This implies that

$$\cos(\sqrt{z}) = C \cdot \prod_{n=1}^{\infty} \left(1 - \frac{4z}{\pi^2(2n-1)^2}\right),$$

which has the shape of the canonical product for the sequence $\{\frac{4}{\pi^2(2n-1)^2}\}_{n\in\mathbb{N}}$, hence has genus $\boxed{0}$. □